Height Growth Equations for CONUS: Loblolly Pine

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Data

We took the FIA height growth data subset for loblolly pine, filtered out stray (likely keypunch errors) states (IL and MI), FIA plot locations with missing EPA L3 Codes, and where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. We also dropped observations where `EMT` was less than -25 $^{\circ}C$ because their height growth appeared to be too high. This left 319199 growth observations.

The observations are distributed among L3 Codes as follows:

NA_L3NAME	NA_L3CODE	n
Southeastern Plains	8.3.5	108733
Piedmont	8.3.4	69833
South Central Plains	8.3.7	66960
Middle Atlantic Coastal Plain	8.5.1	30361
Mississippi Valley Loess Plains	8.3.6	9555
Southern Coastal Plain	8.5.3	9381
Ouachita Mountains	8.4.8	6249
Ridge and Valley	8.4.1	6123
Southwestern Appalachians	8.4.9	4747
Interior Plateau	8.3.3	1736
Arkansas Valley	8.4.7	1455
Blue Ridge	8.4.4	1179
Western Gulf Coastal Plain	9.5.1	764
Mississippi Alluvial Plain	8.5.2	660
East Central Texas Plains	8.3.8	570
Boston Mountains	8.4.6	300
Interior River Valleys and Hills	8.3.2	182
Texas Blackland Prairies	9.4.7	107

NA_L3NAME	NA_L3CODE	n
Ozark Highlands	8.4.5	96
NA	NA	88
Northern Piedmont	8.3.1	79
Central Appalachians	8.4.2	25
Atlantic Coastal Pine Barrens	8.5.4	10
Cross Timbers	9.4.5	6

Height Growth Prediction

We fit the equation:

$$\Delta ht = ht_{max}\beta_1\beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl + \beta_5 PPT_{at} + \beta_6 PPT_{sm} + \beta_7 EMT)}(1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \tag{1}$$

to the all 319199 growth observations, estimating β_0 through β_6. Where:

- ht = total height (feet)
- cr = crown ratio (fraction of total height)
- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- PPT_{at} = autumn precipitation (mm)
- PPT_{sm} = summer precipitation (mm)
- EMT = extreme minimum temperature ($^\circ C$)

ht_{max} was estimated by finding the maximum height observed in the data set for loblolly pine. In our data that value is: 152 feet.

The parameter estimates and fit statistics are:

	Coef.	Std. error	t-stat.	p
b1	0.0201587	0.0001274	158.27014	0
b2	1.6947807	0.0117088	144.74407	0
b3	0.3043435	0.0036709	82.90778	0
b4	0.0009686	0.0000184	52.52240	0
b5	0.0007283	0.0000220	33.11729	0
b6	0.0011422	0.0000138	83.03554	0
b7	-0.0164485	0.0002725	-60.36997	0

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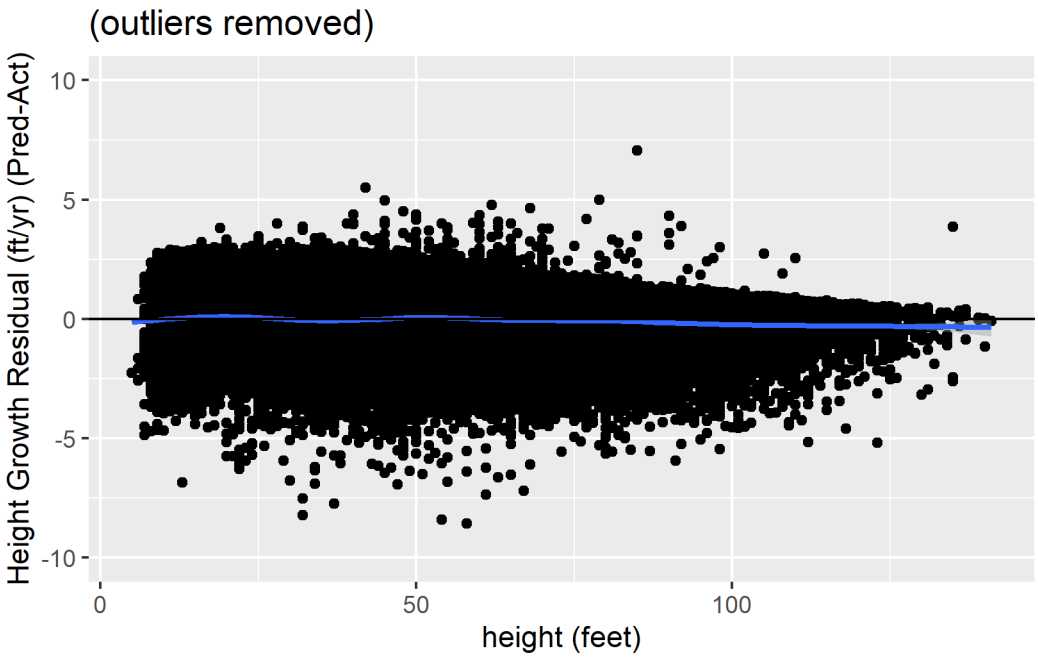
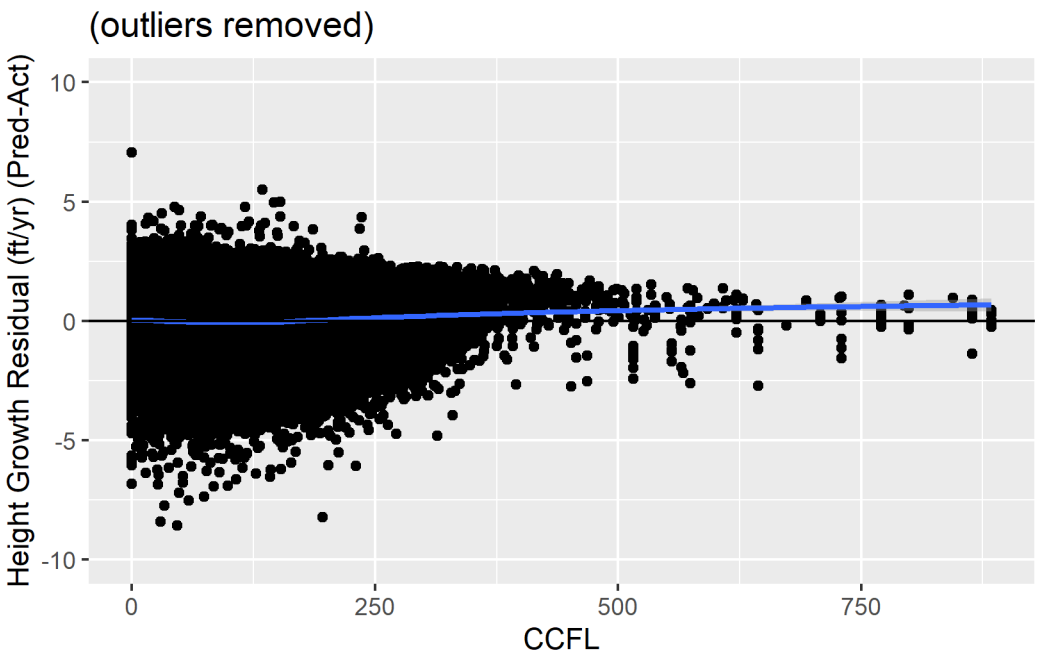
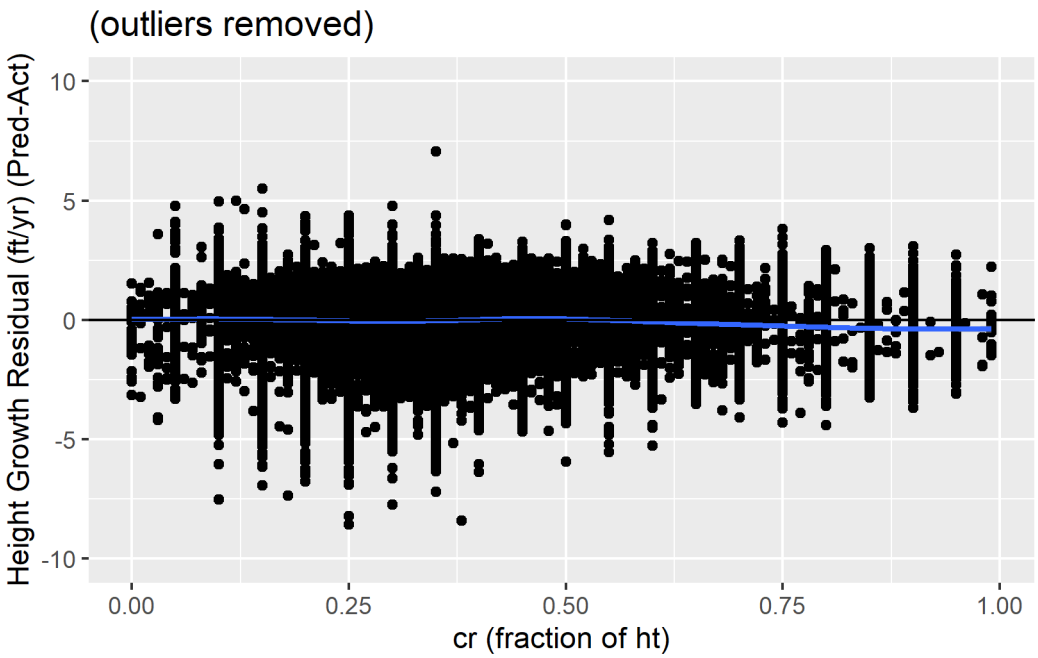
Residual Standard Error: 5.95708824216551 on 319192 degrees of freedom. AIC =  2045130
```

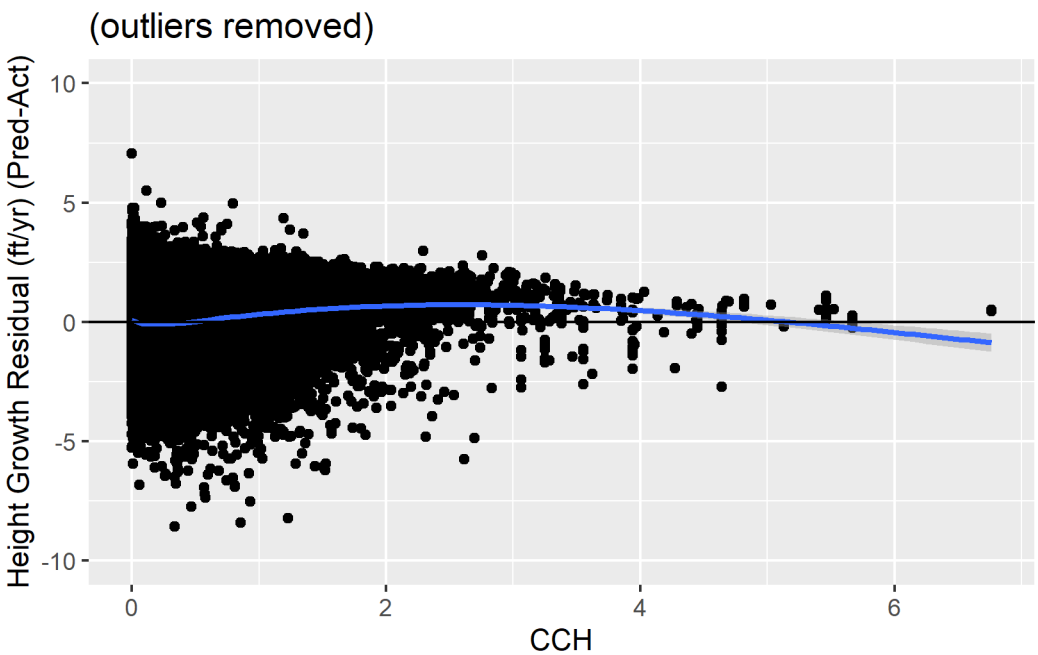
Residual Analysis for Equation 1

Mensurational Variables






(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0.00
0.25
0.50
0.75
1.00
cr (fraction of ht)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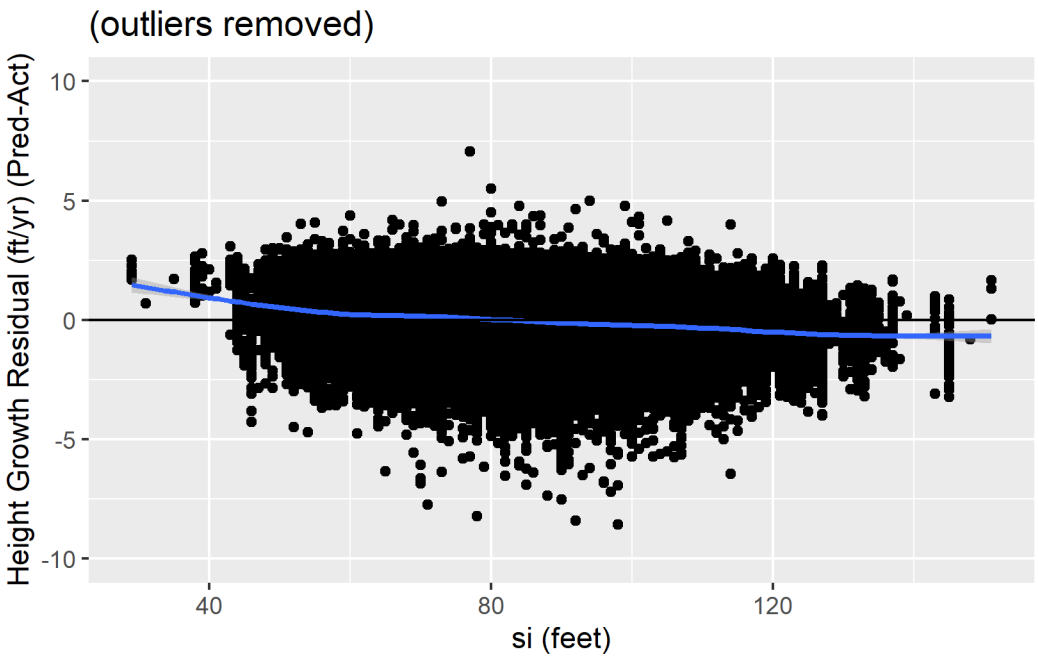
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
2
4
6
CCH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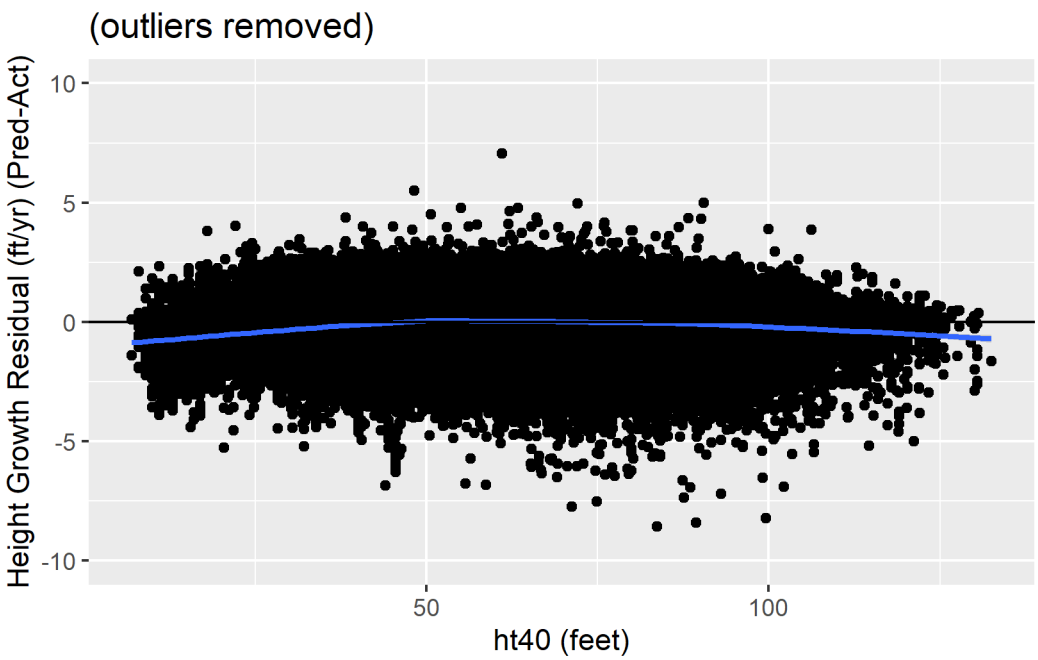
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
40
80
120
si (feet)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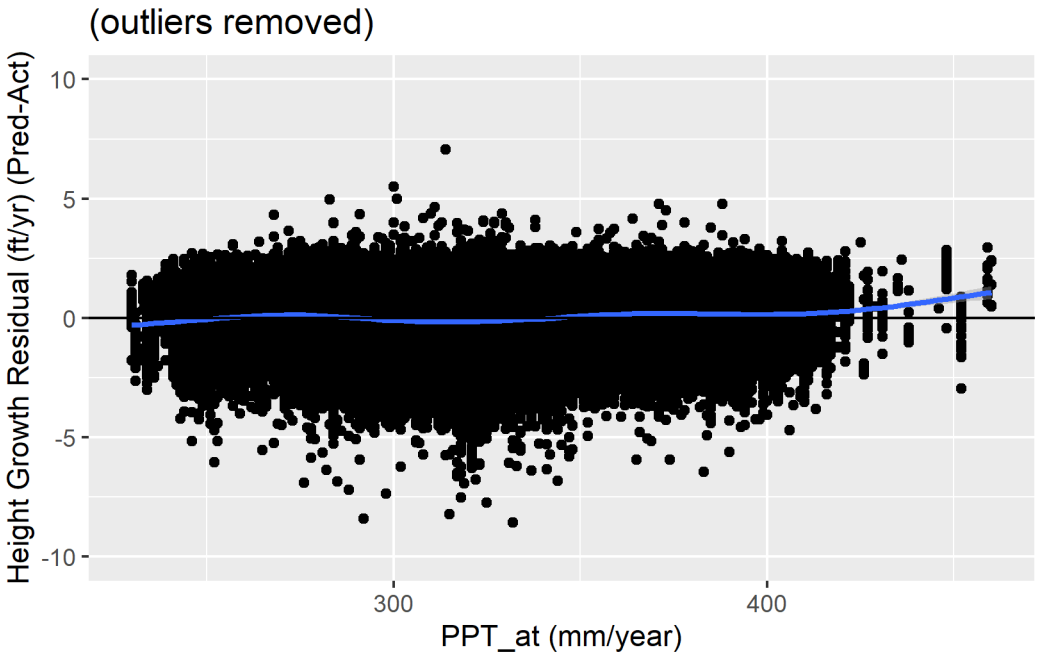
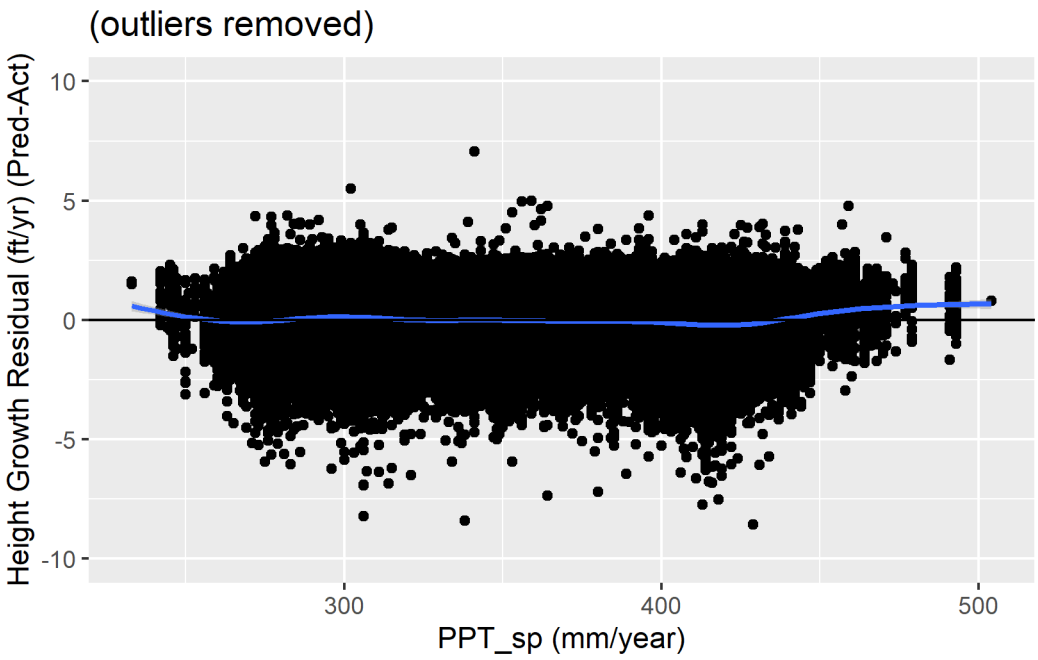
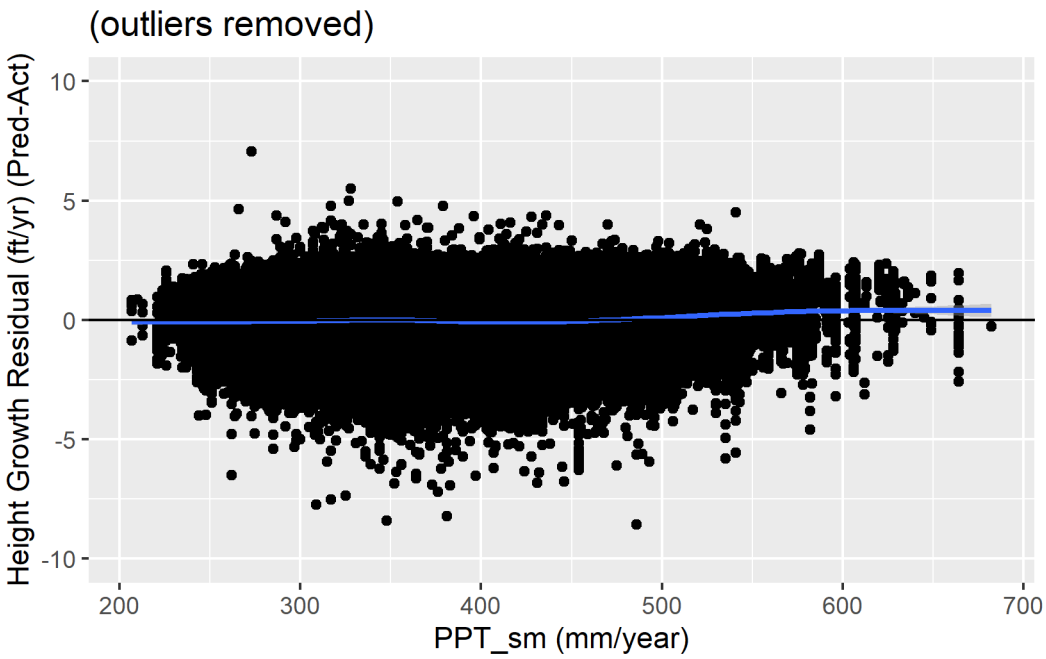
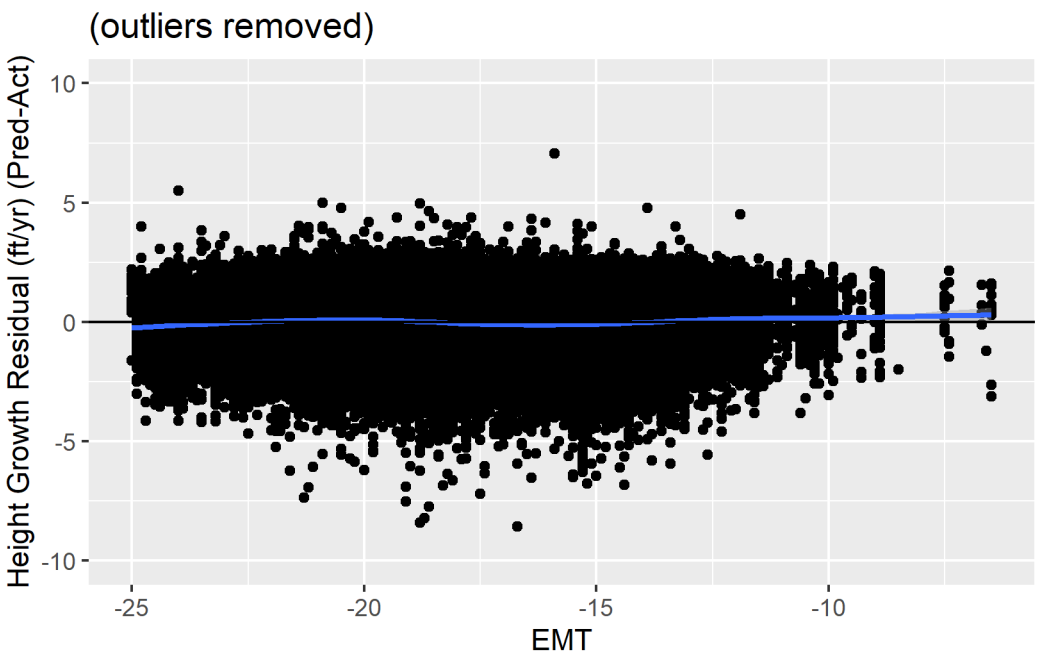
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
50
100
ht40 (feet)

Environmental and Climate Variables






(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
200
300
400
500
600
700
PPT_sm (mm/year)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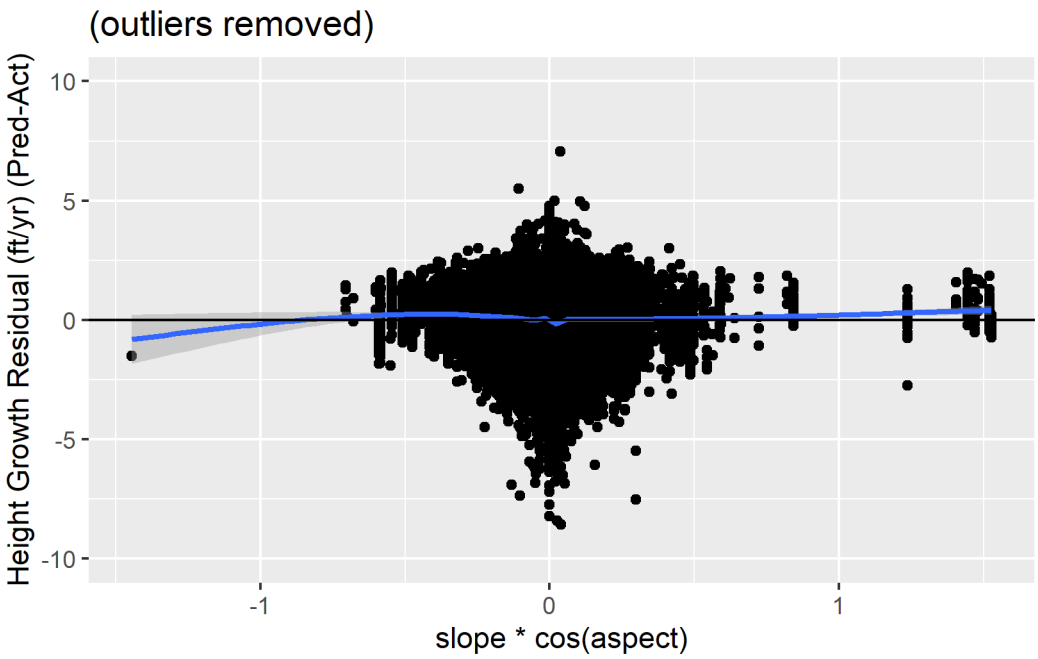
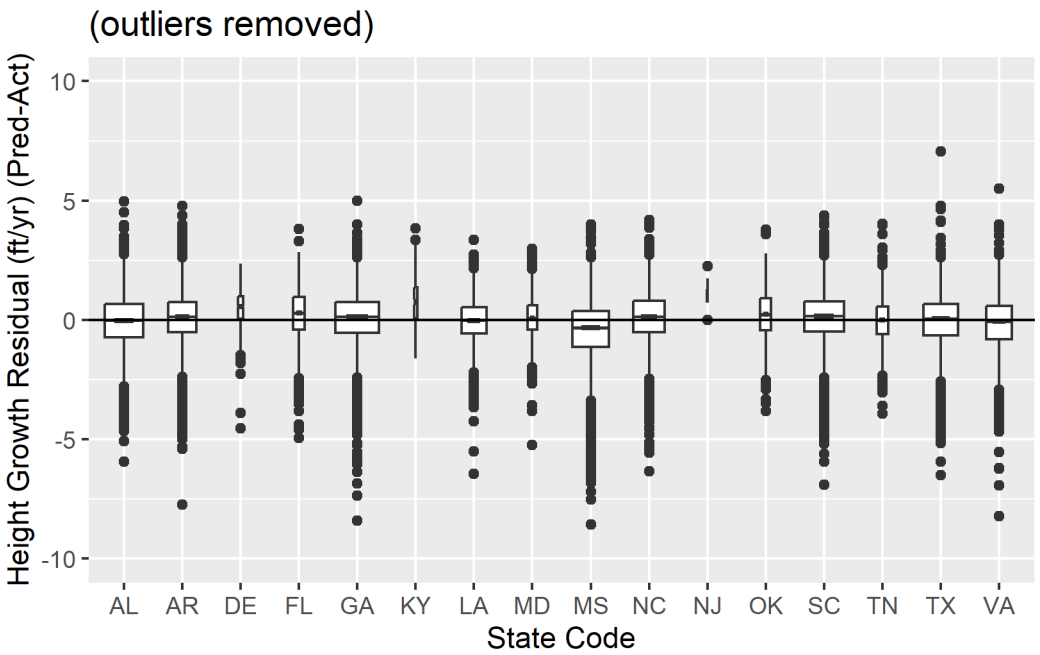
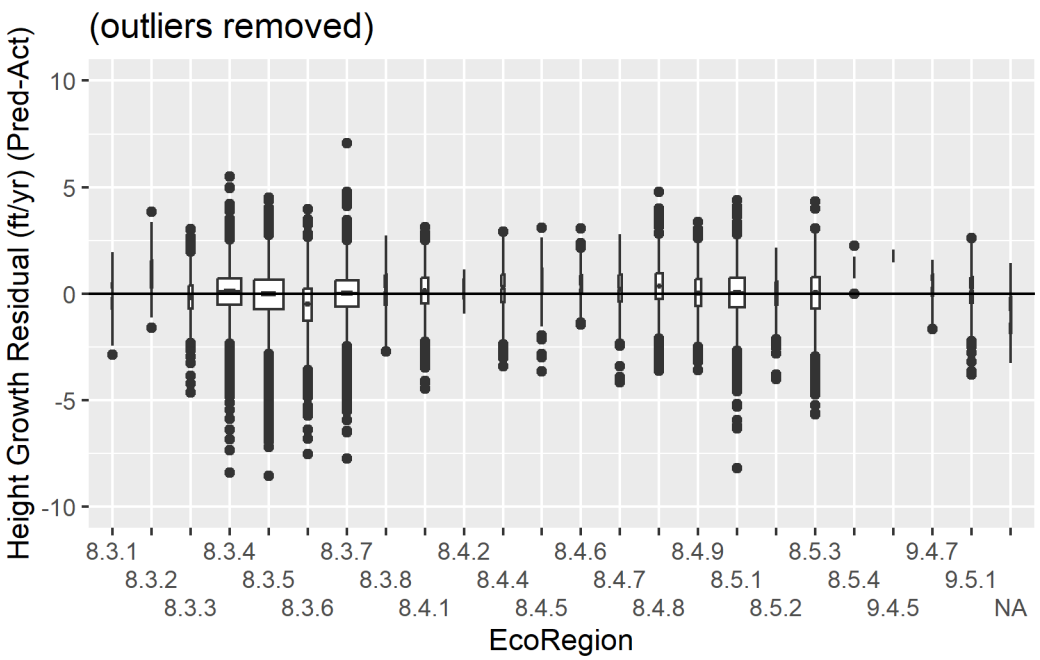
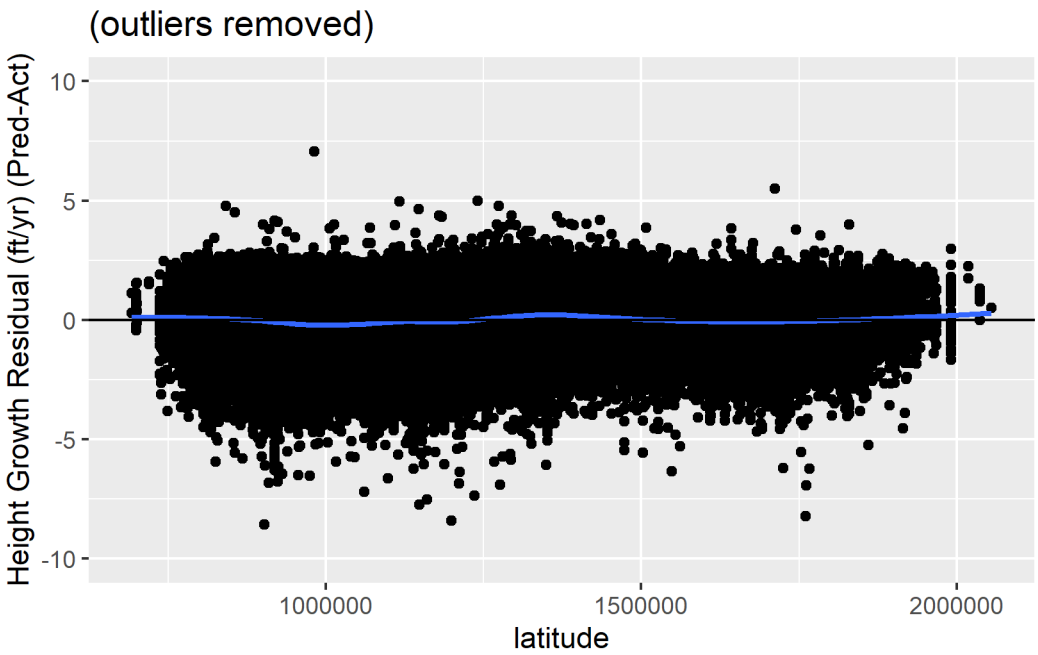
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-25
-20
-15
-10
EMT

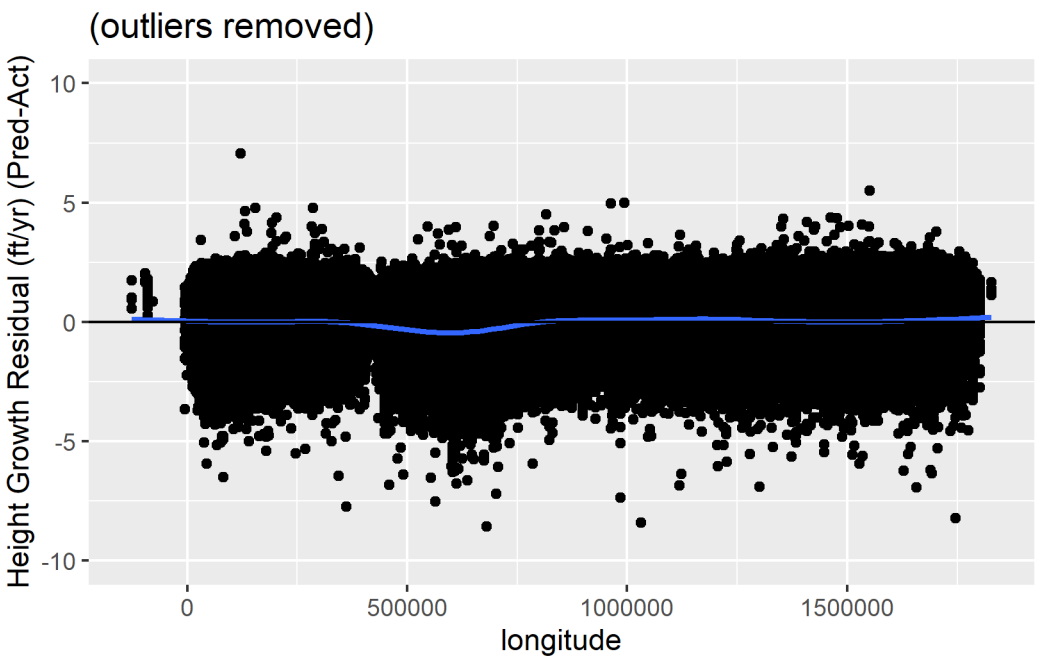

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-1
0
1
slope * cos(aspect)

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-1.0
-0.5
0.0
0.5
1.0
slope * sin(aspect)


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
AL
AR
DE
FL
GA
KY
LA
MD
MS
NC
NJ
OK
SC
TN
TX
VA
State Code


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
8.3.1
8.3.2
8.3.3
8.3.4
8.3.5
8.3.6
8.3.7
8.3.8
8.4.1
8.4.2
8.4.4
8.4.5
8.4.6
8.4.7
8.4.8
8.4.9
8.5.1
8.5.2
8.5.3
8.5.4
9.4.5
9.4.7
9.5.1
NA
EcoRegion


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
1000000
1500000
2000000
latitude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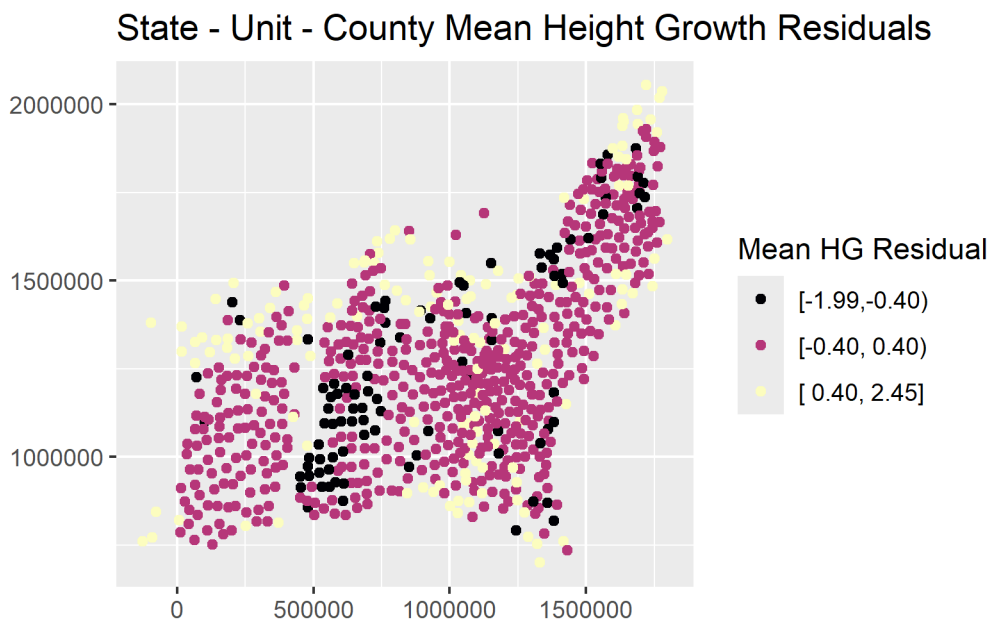
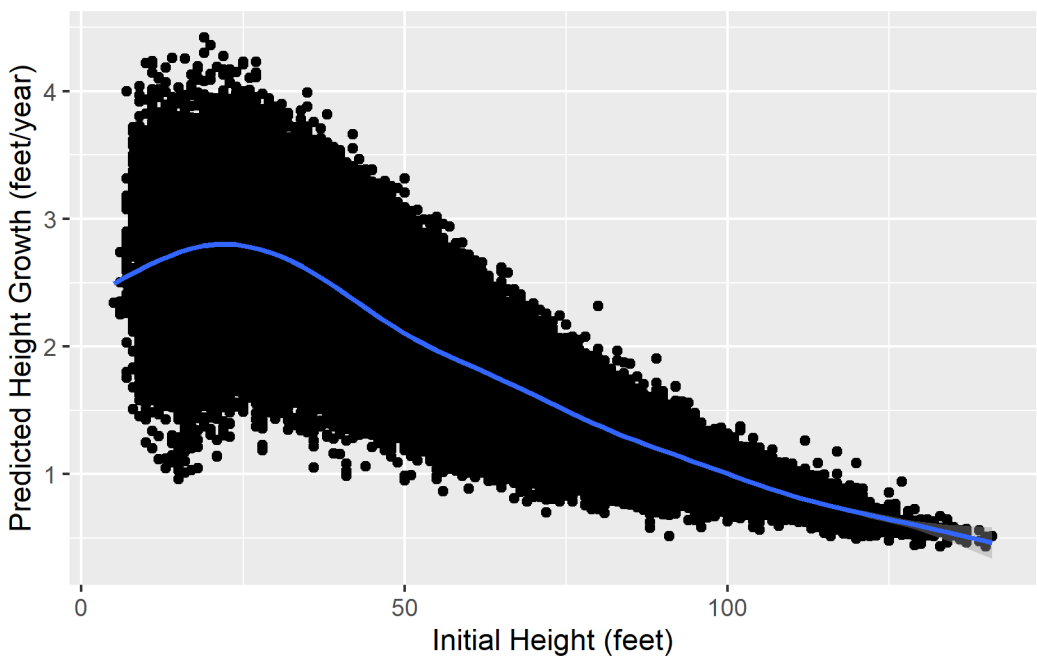
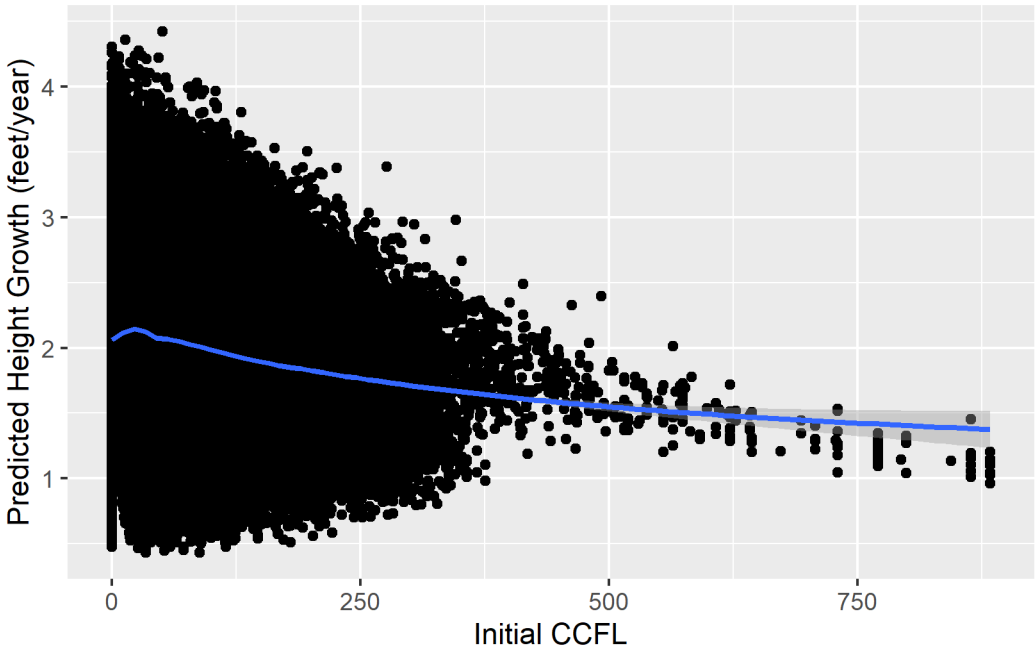
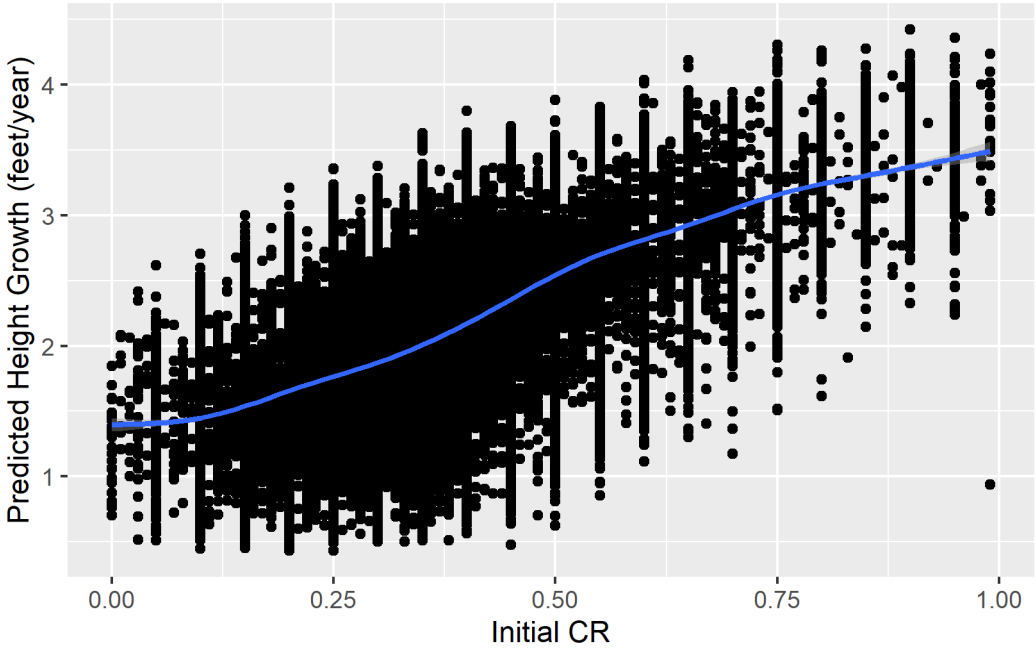
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
500000
1000000
1500000
longitude

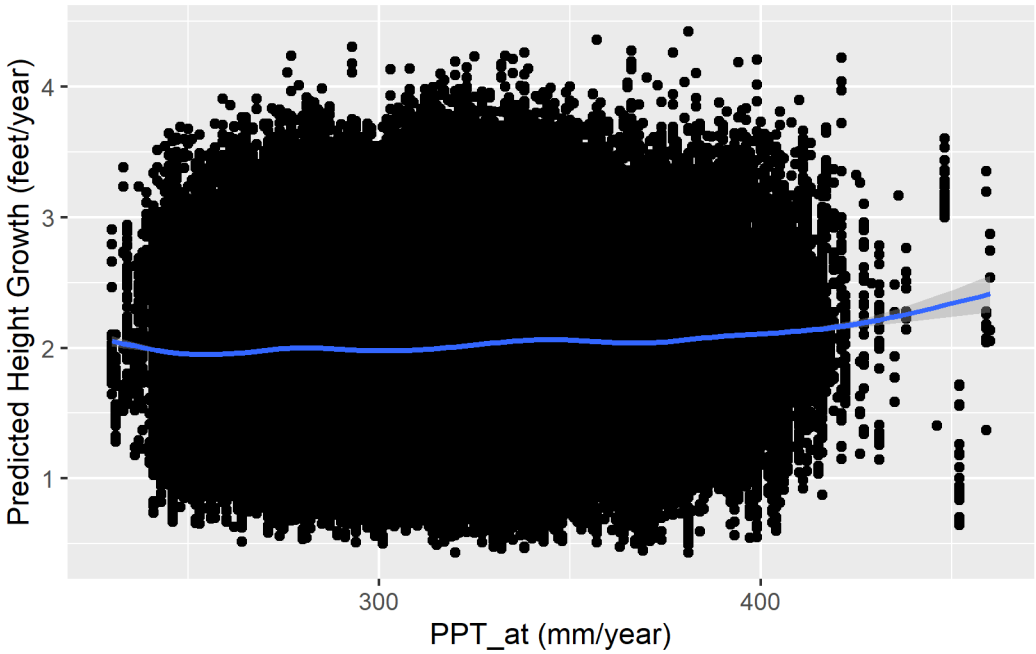


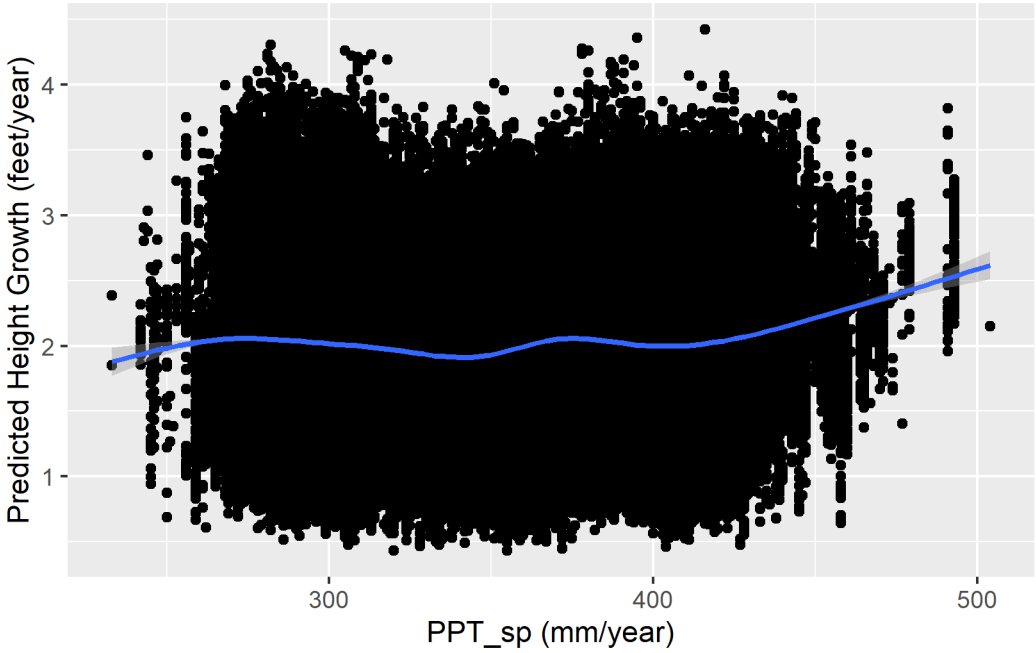
Prediction Performance

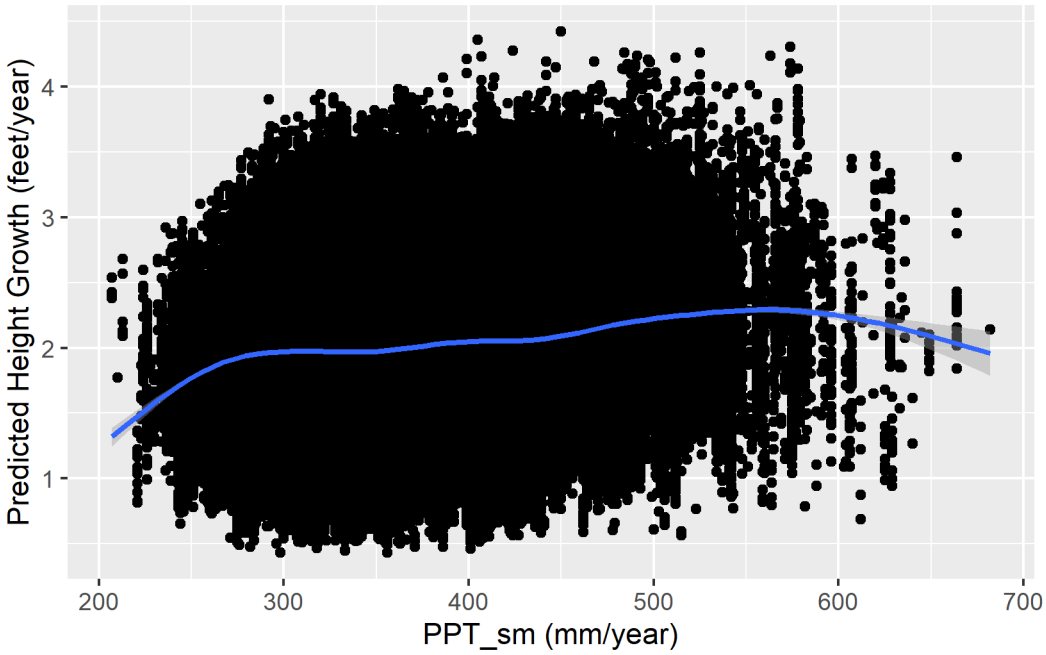


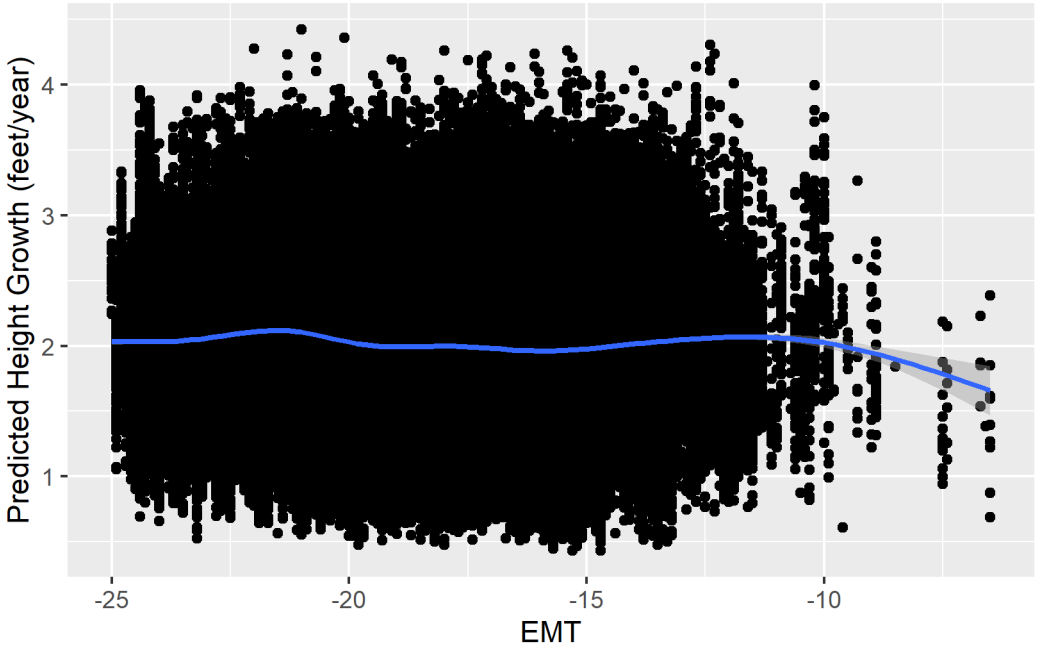

Predicted Height Growth (feet/year)
4
3
2
1
0
250
500
750
Initial CCFL


Predicted Height Growth (feet/year)
4
3
2
1
0.00
0.25
0.50
0.75
1.00
Initial CR


Predicted Height Growth (feet/year)
4
3
2
1
300
400
PPT_at (mm/year)


Predicted Height Growth (feet/year)
4
3
2
1
300
400
500
PPT_sp (mm/year)


Predicted Height Growth (feet/year)
4
3
2
1
200
300
400
500
600
700
PPT_sm (mm/year)


Predicted Height Growth (feet/year)
4
3
2
1
-25
-20
-15
-10
EMT